



# Government-University Identity Management Opportunities

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# ***Current State of Affairs (60 years old now)***

- You apply to the application owner for a password
- You use the password to access the system
- You forget the password
- The application owner gives you a new password
- You use the new password to access the system
- You forget the password
- <infinite do loop>
- No identity proofing
- No way to know who is actually on the system (Your secretary? Your postdoc? Your dog? Osama?)

# ***Foundational Assumption***

- *Government online services shall trust externally-issued electronic identity credentials at known levels of assurance (LOA)*
- *Online applications shall determine required credential LOA using a standard methodology based on:*
  1. *Risk assessment using standard tool,*
  2. *OMB M-04-04 determines required authN LOA*
  3. *NIST SP 800-63 translates required LOA to credential technology*

# ***E-Authentication LOA and What They Mean***

Level 1

- Little or no assurance of identity; assertion-based identity authentication

Level 2

- Some assurance of identity; assertion-based identity authentication or policy-thin PKI

Level 3

- Substantial assurance of identity; cryptographically-based identity authentication

Level 4

- High assurance of identity; cryptographically-based identity authentication

# ***E-Authentication LOA and What They Service\*\****

Level 1

Level 2

Level 3

Level 4

- Online applications with little or no risk of harm from fraud, hacking; low risk
- Online applications with risk of some harm from fraud, hacking; some risks
- Online applications where there is risk of significant harm from fraud, hacking; significant risks
- Online applications where there is risk of substantial harm from fraud, hacking; substantial risks

# **General Considerations for Determining LOA of an Electronic Identity Credential (EIC)**

- **Identity Proofing** – *how sure are you that the person is who he or she claims to be?*
- **Identity Binding** – *how sure are you that the person proffering the EIC is the person to whom the credential was issued?*
- **Credential integrity** – *how well does the technology and its implementation resist hacking, fraud, etc.?*

# ***Summary of Lower-Level Identity Credentials***

- **Level 1:** *UserID/Password, SAML assertion (XML text)*
- **Level 2:** *“High entropy” UserID/Password; “policy-lite” PKI, e.g., Fed PKI Citizen and Commerce Class & Federal PKI Rudimentary, TAGPMA Classic Plus (in development)*

# ***Summary of Cryptographic-Based Identity Credentials***

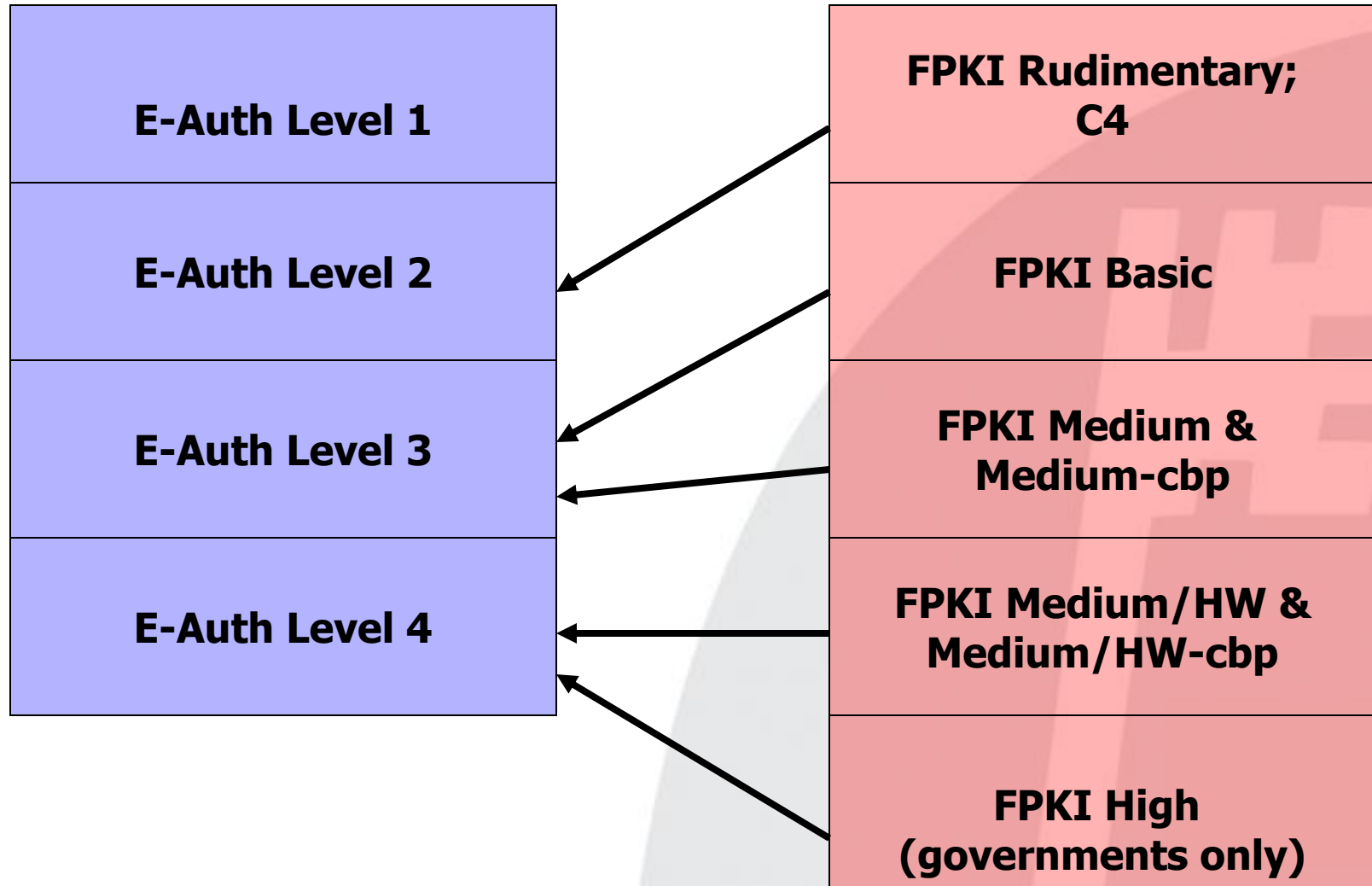
- **Level 3:** *One-time Password; Substantial assurance PKI at FPKI Basic, Medium*
- **Level 4:** *High assurance PKI at FPKI Medium Hardware, High*



# ***A Little Complication***

- The government has TWO LOA classifications:
  1. Federal PKI LOA codified in the Certificate Policies of the Federal PKI Policy Authority
  2. E-Authentication LOA codified in OMB M-04-04

# ***LOA Mapping E-Auth to Fed PKI***



# ***How Can A School Credential be Trusted and Used by a Government Application?***

- **Preferred path** - School joins an Identity Federation that has policy and technology interoperability agreements with E-Authentication Federation
- **Requirements** – Each Federation agrees to ensure its members operate according to minimum requirements for members of the other's Federation.
  - *Presumes substantive policy, technology and management commonality*
- **Alternate path 1** – School becomes a member Credential Service Provider (CSP) of the E-Authentication Federation directly, signing on to the Federation's technology, business, operating and legal requirements
- **Alternate path 2** – One-to-one relationships (sssh!)

# ***US Government E-Authentication Interfederation Model***



# ***E-Authentication Federation Membership Requirements***

- Credential service providers [CSP] submit to credential assessment and evaluation of LOA
- Both Application providers and CSPs sign on to Federation business and operating “standards,” legal agreements.

# ***E-Authentication – InCommon Interfederation Status***

- Candor requires this disclaimer: they're still trying to figure it out after two years and two tries
- Current status: Said to be getting close with inCommon. Policy-grounded MOA on the table and molding; technical interoperability targeted for SAML 2.0; USPerson profile analog of eduPerson profile in 2.0 version but still pretty much generic.
- inCommon needs to up its policy, procedures, documentation and audit requirements to play long-term
- E-Authentication needs to up its privacy protections

# ***Fed PKI “Interfederation” Model***





# ***Fed PKI Cross Certification Process***

- ***Application - LOA?***
- ***Policy Mapping***
  - Mapping Matrices online
  - Cert Policy WG mapping review
  - Collegial back and forth discussions
- ***Technical Interoperability Testing***
  - Testing Protocol online
  - Directory and profiles tested (LDAP and/or X.500)
- ***Review of summary of independent audit results***
  - Map CP → CPS and CPS → PKI Operations
  - Independent auditors, not FPKI auditors
- ***Whole process laid out in “Criteria & Methodology” document online***





# ***FPKI Does Interfederate***

- Cross-certified (test) with Higher Education PKI Bridge, 01/2002
- Cross-certified (production) with CertiPath – Aerospace Industry PKI Bridge at Medium Hardware (EAuth Level 4), 07/2006
- Cross-certification under way with SAFE-Biopharma PKI Bridge, Medium and Medium Hardware
- Processes and procedures spelled out in “Criteria and Methodology” Document online



# ***Current Model for Assertion-Based Interfederation (still a work in progress)***

- For E-Authentication Federation and InCommon to interfederate at LOA 1 (!), E-Auth is requiring InC to:
  - *upgrade its policy, audit and management infrastructure to comply with EAuth model, e.g., compliance with Credential Assessment Framework for LOA 1, signing Business and Operating Standards and sign Legal Agreement*
  - *Satisfy technical interoperability testing using SAML 1.0 technology, though work is proceeding to migrate to SAML 2.0 technology*
- InCommon has designated this state of operation “InCommon Bronze”
- Compatibility with E-Authentication requirements for LOA 2 is called “InCommon Silver”

# ***What about Level 3 Government Apps Today?***

- Universities issuing PKI-based electronic credentials may cross-certify with the Federal PKI at Basic Assurance or above
  - *Ex. MIT Lincoln Lab, University of Texas System (in process)*
  - *Usually 3 – 6 months*
- Or acquire digital certificates from a vendor currently cross-certified with the Federal Bridge at Basic assurance or above
  - *Many options*

# ***Seeded Questions***

- Q: What about the recent DOD notice that PKI individual certs are required for access to their web resources. Will DOD sites ever trust university-issued certificates for access or will we have to shell out \$\$ to get to them?
- A: If you think YOU are fussing about this, imagine how furious State Dep't. and NASA are. Imagine how furious their contractors are.
- A: We (Fed PKI) have been talking to DOD about this issue and we hope to see progress in 2007.
- Q: What the heck is “password entropy?”
- A: See next slide

# ***Password Entropy (Copied From Bill Burr)\****

- Entropy is measure of randomness in a password
  - *Stated in bits, a password with 24 bits of entropy is as hard to guess as a 24 bit random number*
  - *The more entropy required in the password, the more trials the system can allow*
- It's easy to calculate the entropy of a system-generated, random password
  - But users can't remember these*
- Much harder to estimate the entropy of user-chosen passwords
  - *Composition rules and dictionary rules may increase entropy*

**\*NIST KBA Symposium, Feb. 9, 2004**

# Resources

- [www.cio.gov/eauthentication](http://www.cio.gov/eauthentication)
- [www.cio.gov/fpkipa](http://www.cio.gov/fpkipa)
- <http://csrc.nist.gov>
- [www.cio.gov/ficc](http://www.cio.gov/ficc)

# Notes

- 1.) How does your institution issue IDs? Is there a signature station requiring photo ID? What about non-resident scientists from Japan?
- 2.) What is password entropy? Complex vs. long passwords. How often must they be changed? Can they be re-used? Pass dictionary test?
- 3.) How secure is the authentication infrastructure? Can one person jeopardize the process? How are ID records stored?
- 4.) How is the process documented? Are there templates to use as a framework that will pass Federal standards?